AI Module 1 QB solved

1. Define Artificial Intelligence. Discuss the four aspects of AI. (8 Marks)

Definition of Artificial Intelligence

Artificial Intelligence (AI) is a branch of computer science that deals with developing intelligent machines capable of performing tasks that normally require human intelligence, such as thinking, learning, reasoning, problem-solving and decision-making.

Four Aspects of Artificial Intelligence

1. Thinking Humanly (Cognitive Modelling Approach)

- This approach aims to make computers think like human beings.
- It involves understanding human thinking, learning, memory and problem-solving.
- Human thinking can be studied through introspection, psychological experiments and brain imaging.
- Example: A computer program that solves problems using reasoning similar to humans.

2. Thinking Rationally (Laws of Thought Approach)

- This approach focuses on making computers reason correctly using logical principles.
- It uses formal logic and rules of correct reasoning to derive conclusions.
- Correct conclusions are obtained when valid reasoning is applied to correct premises.
- Example: A computer program that uses logical rules to solve mathematical problems.

3. Acting Humanly (Turing Test Approach)

- This approach aims to develop machines that behave like human beings.
- Alan Turing proposed the Turing Test in 1950 to evaluate intelligent behaviour.
- It requires capabilities such as natural language processing, knowledge representation, automated reasoning and machine learning.
- Example: A chatbot that communicates with humans in a human-like manner.

4. Acting Rationally (Rational Agent Approach)

- This approach focuses on developing intelligent agents that choose appropriate actions to achieve the best expected outcome.
- A rational agent makes decisions based on available information, prior knowledge, possible actions and its performance measure.
- It aims to achieve goals rather than simply imitate human behaviour.

- Example: A self-driving car selecting a safe and efficient route based on traffic conditions.

Conclusion

The four aspects of AI are thinking humanly, thinking rationally, acting humanly and acting rationally. These approaches help in understanding and developing intelligent systems capable of reasoning, learning and performing tasks effectively.

2. Discuss the state-of-the-art applications of Artificial Intelligence. (8 Marks)

Introduction

Artificial Intelligence (AI) is used in various fields to develop intelligent systems that can learn, reason, solve problems, make decisions and perform tasks automatically.

Applications of Artificial Intelligence

1. Robotic Vehicles

AI is used in autonomous vehicles to perceive their surroundings, navigate roads, detect obstacles and make driving decisions with minimal human intervention.

2. Speech Recognition

AI enables computers to recognize and interpret human speech. It is used in voice assistants such as Siri and Alexa to understand voice commands and respond to users.

3. Autonomous Planning and Scheduling

AI helps systems plan activities, allocate resources and schedule tasks efficiently. It is used in transportation, manufacturing and project management.

4. Game Playing

AI is used to develop programs that play games such as chess. These systems evaluate possible moves and select suitable strategies to improve their chances of winning.

5. Spam Filtering

- AI is used to identify and filter unwanted or suspicious emails.
- It analyses email content and patterns to distinguish spam messages from genuine messages.
- Example: Email services automatically moving spam messages into the spam folder.

6. Logistics

- AI helps in planning transportation routes, managing inventory and scheduling deliveries.
- It improves efficiency by reducing delivery time and operational costs.
- Example: A delivery system selecting an efficient route for transporting goods.

7. Machine Translation

- AI translates text or speech from one language into another.
- It analyses language patterns and context to produce meaningful translations.
- Example: Google Translate translating Kannada into English.

Conclusion

AI has important applications in robotics, speech recognition, planning, gaming, spam filtering, logistics and machine translation. These applications improve efficiency, automate tasks and support better decision-making.

3. Is AI a science, engineering, neither or both? Explain. (8 Marks)

Introduction

Artificial Intelligence (AI) is both a science and an engineering discipline. It involves studying intelligent behaviour and developing computer systems that can think, learn, reason and act intelligently.

AI as a Science

- AI is a science because it studies the principles and mechanisms of intelligence.
- It investigates how humans think, learn, reason and solve problems.
- It develops theories and models to understand intelligent behaviour.
- It uses experiments and observations to test these theories.
- Example: Studying how humans solve problems and developing computational models of their reasoning.

AI as Engineering

- AI is an engineering discipline because it involves designing and developing intelligent systems to solve real-world problems.
- It applies algorithms, programming techniques and computational models to build practical applications.

- It focuses on creating systems that perform tasks efficiently and achieve specific goals.
- Example: Developing self-driving cars, speech recognition systems and intelligent robots.

Why AI Is Both Science and Engineering

- Science helps us understand the principles of intelligence.
- Engineering uses those principles to design and build intelligent systems.
- Scientific research provides the knowledge, while engineering converts that knowledge into practical applications.
- Both contribute to the continuous development and improvement of AI systems.

Conclusion

AI is both a science and an engineering discipline. As a science, it studies intelligence and its underlying principles; as engineering, it applies that knowledge to develop intelligent machines and solve real-world problems.

4. Examine whether the following tasks can currently be solved by computers.

i) Playing a decent game of table tennis (ping-pong)

Yes, to a certain extent. Robots can play table tennis by using cameras, sensors, and control systems to track the ball and move the paddle. However, playing consistently well against skilled human players remains challenging because the ball moves rapidly and its direction can change unexpectedly.

ii) Discovering and proving new mathematical theorems

Yes. Computers can prove mathematical theorems using automated reasoning and formal logic. They can also help discover new results by exploring mathematical patterns. However, discovering and proving arbitrary new theorems remains difficult and may require human guidance.

iii) Giving competent legal advice in a specialized area of law

Yes, to a limited extent. AI systems can analyze legal documents, search legal databases, and provide information about specific areas of law. However, they may

misunderstand complex situations or miss important details. Therefore, expert human supervision is still important.

iv) Performing complex surgical operations

Yes, with limitations. Computer-controlled surgical robots can assist surgeons in performing complex operations with precision. However, they are generally operated or supervised by trained medical professionals and do not independently handle every complex surgical situation.

Conclusion

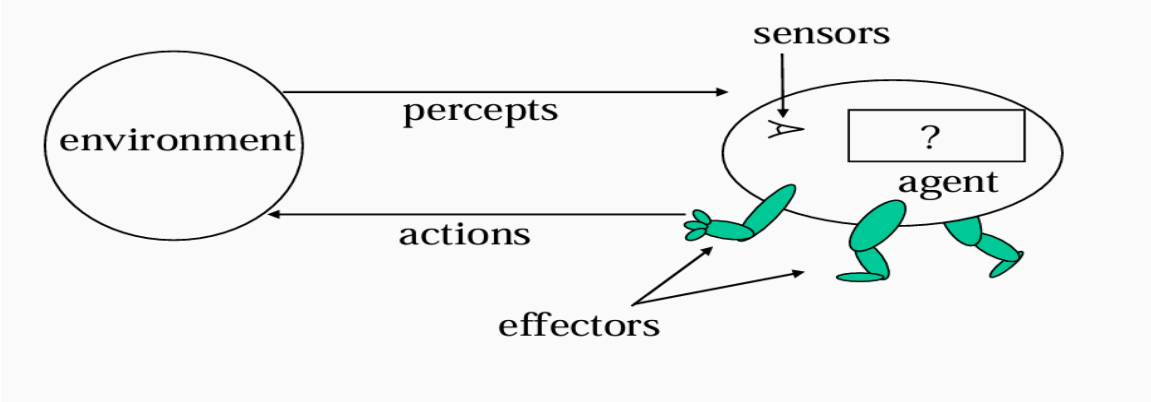
Computers can perform all four tasks to some extent, but their capabilities vary. They are effective in well-defined tasks, while tasks requiring adaptability, judgment, and handling unexpected situations still present challenges.

5. What are agents? Define percepts, actions, goals and environment with reference to the following.

Definition of an Agent

An agent is a system that perceives its environment through sensors and acts upon that environment using actuators (effectors). It receives percepts and selects suitable actions to achieve its goals.

- Percepts: Information received by an agent through its sensors.
- Actions: Operations performed by an agent using its actuators.
- Goals: The objectives the agent tries to achieve.
- Environment: The surroundings in which the agent operates.



a) Automated taxi driving system

- Percepts: Video, sonar, speedometer, odometer, engine sensors, keyboard input, microphone and GPS.
- Actions: Steering, accelerating, braking, using the horn and displaying or speaking information.
- Goals: Maintain safety, reach the destination, obey traffic laws, provide passenger comfort and maximize profit.
- Environment: Urban streets, freeways, traffic, pedestrians, weather conditions and customers.

b) Medical diagnosis system

- Percepts: Patient symptoms, medical findings and the patient's answers.
- Actions: Asking questions, conducting tests and recommending treatments.
- Goals: Keep patients healthy and minimize costs.
- Environment: Patients and hospitals.

c) Satellite image analysis system

- Percepts: Pixels of varying intensity and colour in satellite images.
- Actions: Printing or producing a categorization of the scene.
- Goals: Correctly categorize the scene shown in the image.
- Environment: Images obtained from orbiting satellites.

d) Part-picking robot

- Percepts: Pixels of varying intensity captured by its visual sensors.
- Actions: Picking up parts and sorting them into bins.
- Goals: Place parts in the correct bins.
- Environment: Conveyor belts carrying different parts.

6. What are the characteristics of rational agents?

A rational agent is an agent that selects the action expected to maximize its performance measure based on the information it has received and its built-in knowledge.

Characteristics of Rational Agents

1. Rationality: A rational agent selects the best possible action to achieve its goals based on available information.

2. Perception: It perceives its environment through sensors and uses the information received to make decisions.

3. Goal-oriented behaviour: It performs actions to achieve specific goals and maximize its performance.

4. Autonomy: Its behaviour is determined by its own experience. It can learn and adapt instead of depending entirely on predefined instructions.

5. Reactivity: It responds to changes in the environment by selecting suitable actions.

6. Information gathering: It can perform actions to obtain useful information and improve future decisions.

7. Adaptability and learning: It learns from its own experience and improves its performance over time.

8. Limited knowledge: A rational agent is not omniscient; it does not know every possible outcome. It chooses the best expected action using the information available.

9. Performance-based decision-making: Its rationality depends on four factors:

- Performance measure of success.
- Prior knowledge of the environment.
- Actions available to the agent.
- Percept sequence received so far.

Conclusion: A rational agent perceives its environment, selects actions to maximize expected performance, and uses experience to learn and adapt.

7. Describe PEAS with respect to the following agent types.

PEAS stands for:

- P – Performance Measure: Criteria used to evaluate the agent's success.
- E – Environment: The surroundings in which the agent operates.
- A – Actuators: The mechanisms used by the agent to perform actions.
- S – Sensors: The mechanisms used by the agent to receive information.

The first four examples below follow your PDF's "Examples of agents in different types of applications" table. The remaining examples are completed using the same PEAS framework.

Module 1 notes AI.pdf

PEAS descriptions

Agent type	Performance Measure (P)	Environment (E)	Actuators (A)	Sensors (S)
a) Automated taxi driving	Safety, reaching destination, obeying laws, passenger comfort, maximizing profit	Roads, traffic, pedestrians, weather, passengers	Steering, accelerator, brakes, horn, display/speaker	Cameras, sonar, speedometer, GPS, odometer, engine sensors
b) Medical diagnosis system	Healthy patients, accurate diagnosis, minimizing costs	Patients, hospitals	Ask questions, recommend tests and treatments	Patient symptoms, medical findings, patient answers

c) Satellite image analysis	Correct classification of scenes	Images from orbiting satellites	Produce or print scene classification	Image pixels, colour and intensity
d) Part-picking robot	Place parts in correct bins accurately	Conveyor belts carrying parts	Robotic arm, gripper, motors	Camera and visual sensors detecting parts
e) Playing a tennis match	Win the match, score points, minimize errors	Tennis court, opponent, net, ball and weather	Move, run, swing racket, serve	Cameras, vision sensors, position and motion sensors
f) Performing a high jump	Clear the maximum height without fouling	High-jump track, bar, landing mat and weather	Leg and body movements, running and jumping	Cameras, motion sensors, balance and position sensors
g) Bidding in an auction	Win the desired item at the lowest acceptable price	Auction website or auction room, seller and competing bidders	Submit, increase or withdraw a bid	Current price, competing bids, auction rules and remaining time
h) Playing soccer	Win the match, score goals and prevent the opponent from scoring	Soccer field, ball, teammates, opponents and referee	Run, kick, pass, shoot, tackle and move	Cameras, vision sensors, player positions and ball position

i) Exploring Titan's subsurface oceans	Collect useful scientific data, explore safely and complete the mission	Titan's subsurface ocean, ice, pressure and temperature conditions	Propulsion systems, robotic arms, sampling and communication systems	Cameras, sonar, temperature, pressure and chemical sensors
j) Shopping for used AI books online	Find suitable books at low prices, minimize delivery costs and obtain good-condition books	Online stores, sellers, listings and delivery services	Search, filter, compare, add to cart and purchase	Book descriptions, prices, ratings, availability and seller information
k) Knitting a sweater	Produce a correctly sized, good-quality sweater with minimal material waste	Yarn, needles or knitting machine, sweater pattern and workspace	Move needles, manipulate yarn and form stitches	Cameras, yarn-tension sensors, stitch and position sensors
l) Bidding in an auction	Same as (g): win the desired item within the budget	Same as (g): auction platform, seller and competing bidders	Same as (g): place or increase bids	Same as (g): current price, bids, rules and time remaining

Exam tip: Remember the order P–E–A–S: Performance measure, Environment, Actuators, Sensors. Question (l) repeats question (g), so the same answer applies.

8. Define the following:

i) Artificial Intelligence (AI)

Artificial Intelligence is a branch of computer science that deals with developing intelligent machines or programs capable of performing tasks that normally require human intelligence, such as learning, reasoning, problem-solving and decision-making.

ii) Agent

An agent is anything that perceives its environment through sensors and acts upon that environment through actuators or effectors. It receives percepts and maps them to appropriate actions.

Example: A robotic vacuum cleaner senses dirt and moves to clean it.

iii) Rational Agents

A rational agent is an agent that selects the action expected to maximize its performance measure based on the percept sequence it has received and its built-in knowledge.

Example: A self-driving car selects a safe route while following traffic rules.

iv) Rationality

Rationality is the ability of an agent to choose an appropriate action that maximizes its expected performance based on the available information and its goals.

Rationality depends on four factors:

1. Performance measure of success.
2. Agent’s prior knowledge of the environment.
3. Actions available to the agent.
4. Agent’s percept sequence to date.

Note: Rationality does not mean knowing everything; it means choosing the best expected action using the available information.

v) Logical Reasoning

Logical reasoning is the process of using facts, rules and logical relationships to derive valid conclusions. In Artificial Intelligence, it helps systems infer new information, solve problems and make decisions.

Example: If all humans are mortal and Ravi is a human, then Ravi is mortal.

9. Summarize the properties of task environment? Give examples.

A task environment is the environment in which an intelligent agent operates. Before designing an intelligent agent, we must specify the properties of its task environment.

According to the Module 1 notes, task environments are classified into the following six types.

1. Fully Observable / Partially Observable

- Fully observable: The agent's sensors provide all the information needed to determine the state of the environment.
    - Example: Chess, where the positions of all pieces are visible.
- Partially observable: The agent's sensors do not provide complete information about the environment.
    - Example: Driving in heavy fog or diagnosing a patient using symptoms.

2. Deterministic / Stochastic

- Deterministic: The next state of the environment is completely determined by the current state and the agent's action.
    - Example: A simple puzzle with fixed rules.
- Stochastic: The environment has unpredictable outcomes.
    - Example: Driving in traffic, where other vehicles may behave unpredictably.

3. Episodic / Sequential

- Episodic: Each task is independent, and the current action does not depend on previous episodes.
    - Example: Classifying individual images.
- Sequential: Current actions affect future states and decisions.
    - Example: Chess and driving.

4. Static / Dynamic

- Static: The environment does not change while the agent is thinking.
    - Example: Solving a crossword puzzle.
- Dynamic: The environment can change while the agent is thinking or before it acts.
    - Example: A soccer match.
- Semi-dynamic: The environment itself does not change, but the agent's performance score changes with time.
    - Example: A timed examination.

5. Discrete / Continuous

- Discrete: The environment has a limited number of distinct percepts and actions.
    - Example: Chess, where moves are selected from a set of legal moves.
- Continuous: Percepts and actions can vary smoothly over a range of values.
    - Example: Taxi driving, where speed, steering angle and position vary continuously.

6. Single-agent / Multi-agent

- Single-agent: Only one intelligent agent is involved in performing the task.
    - Example: Solving a crossword puzzle.
- Multi-agent: Two or more intelligent agents interact, and their actions can affect one another. They may be cooperative or competitive.
    - Example: Chess, which involves two competing agents.

10. What are agent programs? Outline the structure of different agent programs listed below and their properties with suitable block diagrams for each.

Definition of Agent Programs

An agent program is a program that implements the agent function. It runs on the agent's physical architecture and maps percepts received from the environment to appropriate actions.

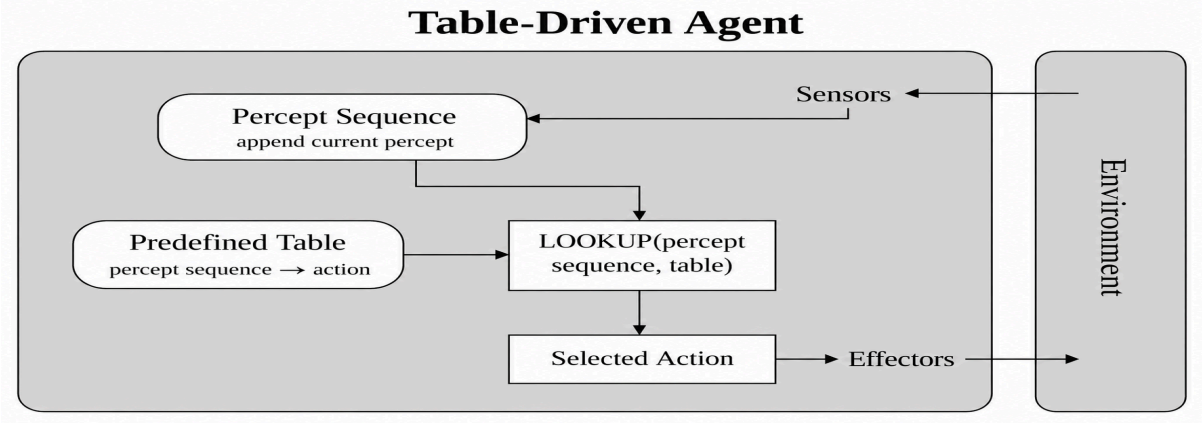
An intelligent agent perceives its environment through sensors and performs actions through actuators. Different agent programs use different methods to select actions.

The five types of agent programs are:

1. Table-driven agent
2. Simple reflex agent
3. Reflex agent with internal state
4. Agent with explicit goals
5. Utility-based agent

(a) Table-Driven Agent

Definition: A table-driven agent selects an action by looking up the complete percept sequence in a predefined table.



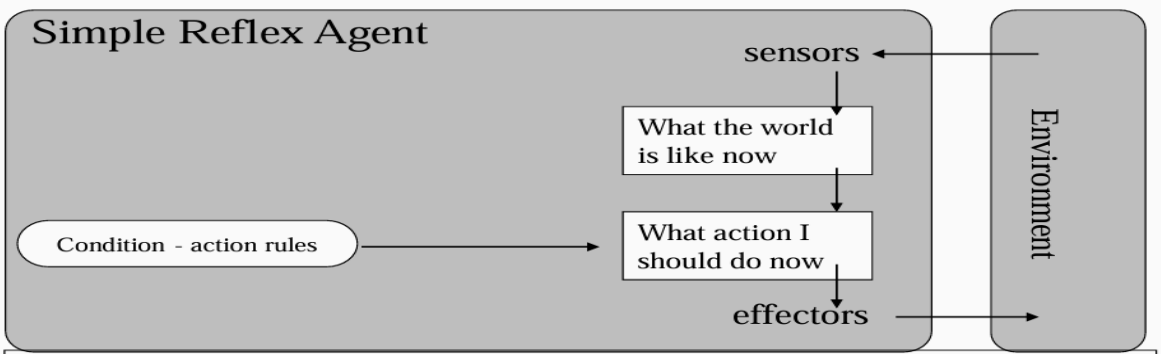
Properties

- Uses a predefined lookup table to select actions.
- Stores percept sequences and their corresponding actions.
- Can select an action only if the relevant sequence is represented in the table.
- Requires a very large table because the number of possible percept sequences can be enormous.
- Difficult to build and maintain for complex environments.

Example: A vacuum-cleaner agent that looks up an action for every possible sequence of locations and dirt conditions.

(b) Simple Reflex Agent

Definition: A simple reflex agent selects an action based only on the current percept, ignoring the history of past percepts. It uses condition–action rules to decide what to do.



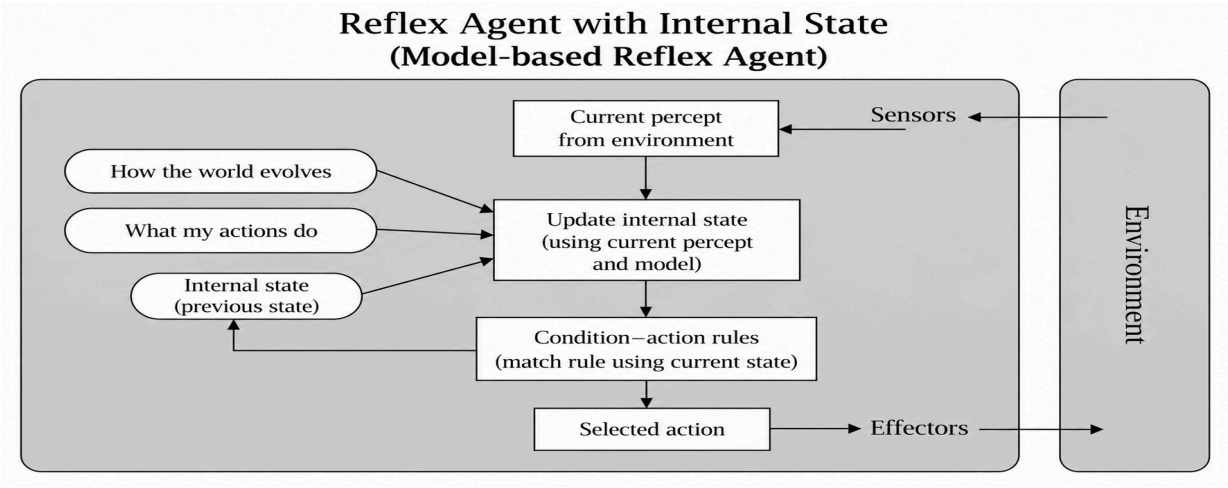
Properties

- Uses condition–action rules: IF condition, THEN action.
- Considers only the current percept.
- Does not maintain memory of past world states.
- Simple and fast to implement.
- Cannot handle situations that require information about hidden states or past events.
- May get stuck in loops or fail when the current percept is insufficient.

Example: An automatic door opens when its sensor detects a person and closes when no person is detected.

(c) Reflex Agent with Internal State

Definition: A reflex agent with internal state, also called a model-based reflex agent, maintains an internal representation of the environment. It updates this state using the current percept and stored information, then selects an action using condition–action rules.



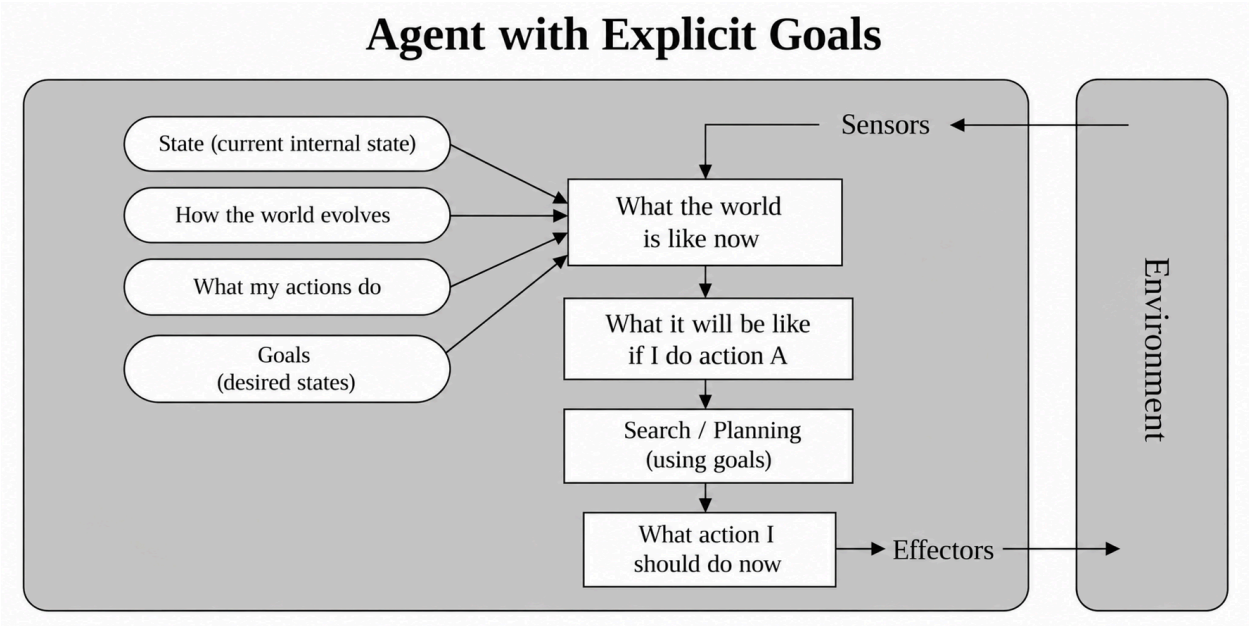
Properties

- Maintains an internal state to remember relevant past percepts.
- Uses a model of how the world changes and how its actions affect the world.
- Suitable for partially observable and dynamic environments.
- Makes better decisions when the current percept alone is insufficient.
- More complex than a simple reflex agent because it must maintain and update its state.

Example: A robotic vacuum cleaner remembers which areas have already been cleaned and moves to another area.

(d) Agent with Explicit Goals

Definition: A goal-based agent selects actions to achieve a specified goal. In addition to information about the current state, it uses goal information to evaluate possible future states and may use search and planning to choose an action.



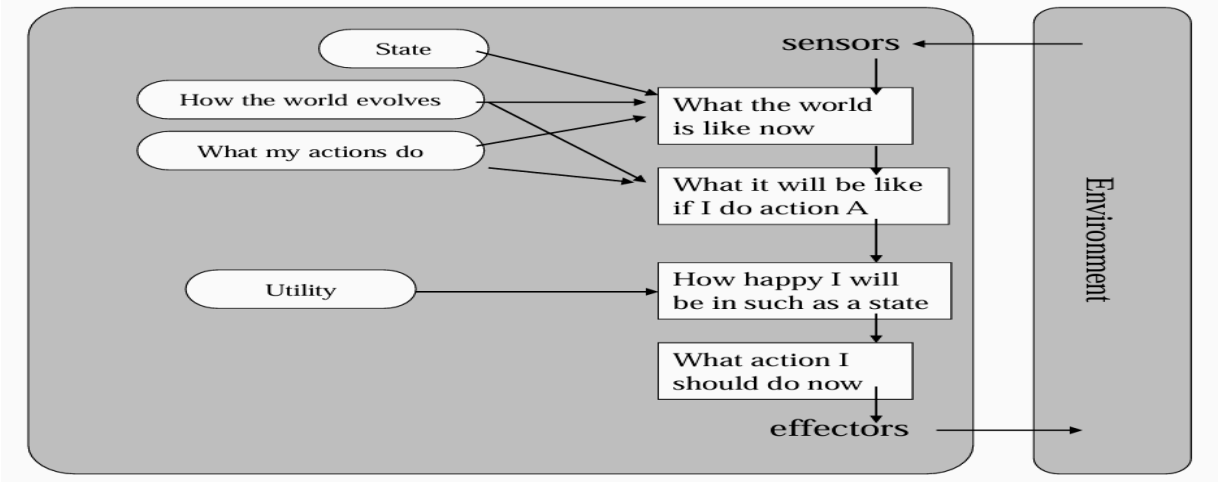
Properties

- Uses explicit goal information to guide action selection.
- Considers future consequences of possible actions.
- Can use search and planning to find a sequence of actions that achieves a goal.
- More flexible than a simple reflex agent because goals can change without rewriting all condition–action rules.
- May require more computation to evaluate different possible actions.

Example: A GPS navigation system plans a route from point A to point B.

(e) Utility-Based Agent

Definition: A utility-based agent selects actions using a utility function, which assigns a numerical value to a state to indicate how desirable or successful it is. It chooses actions that maximize expected utility.



Properties

- Uses a utility function to measure the desirability of states.
- Compares multiple possible alternatives and selects the most beneficial action.
- Can handle situations involving conflicting goals or uncertain outcomes.
- Considers not only whether a goal is achieved, but also how well it is achieved.
- More flexible in decision-making, but requires a suitable utility function and may involve significant computation.

Example: A self-driving car compares possible routes based on travel time, safety and fuel consumption before selecting a route.

Comparison of the five agent programs

Agent program	Main basis for action	Memory / goals / utility
Table-driven	Lookup table of percept sequences	Stores percept history

Simple reflex	Current percept and rules	No memory of past states
Reflex with internal state	Current percept, internal state and rules	Maintains internal state
Explicit goals	Goals and predicted future states	Plans to achieve goals
Utility-based	Expected utility of possible states	Compares how desirable outcomes are

11. Implement a simple reflex agent for the vacuum environment. Run the environment with this agent for all possible initial dirt configurations and agent locations. Record the performance score for each configuration and the overall score.

1. Theory

A simple reflex agent selects an action based only on the current percept, ignoring the history of past percepts. It uses condition–action rules to decide what to do.

The vacuum-cleaner environment contains two rooms, A and B. The agent can perform four actions: Left, Right, Suck and NoOp.

```
function REFLEX-VACUUM-AGENT([location,status]) returns an action
  if status = Dirty then return Suck
  else if location = A then return Right
  else if location = B then return Left
```

Figure 2.8 The agent program for a simple reflex agent in the two-state vacuum environment. This program implements the agent function tabulated in Figure 2.3.

2. Execution table

Each room can be either dirty or clean. Therefore, there are $2^2 = 4$ dirt configurations. Since the agent can start in either A or B, the total number of configurations is $4 \times 2 = 8$.Assumption: The simulation stops when both rooms are clean. This stopping condition requires the simulator to know the state of both rooms; a strict simple reflex agent observing only its current room cannot independently determine that both rooms are clean.

No.	Initial location	Room A	Room B	Actions performed	Score (/2)
1	A	Dirty	Dirty	Suck → Right → Suck	2
2	A	Dirty	Clean	Suck	2
3	A	Clean	Dirty	Right → Suck	2
4	A	Clean	Clean	NoOp	2
5	B	Dirty	Dirty	Suck → Left → Suck	2
6	B	Dirty	Clean	Left → Suck	2
7	B	Clean	Dirty	Suck	2
8	B	Clean	Clean	NoOp	2

3. Performance calculation

The performance measure is one point for each clean room at the end of the simulation.

- Number of configurations = 8
- Maximum score per configuration = 2

- Maximum overall score = $8 \times 2 = 16$
- Overall score obtained = 16/16
- Average score per configuration = 2/2

4. Result

The simple reflex vacuum agent was evaluated for all eight initial configurations. Under the chosen performance measure and stopping condition, it achieved an overall score of 16 out of 16, with both rooms clean at the end of every simulation.

12. Identify the characteristics satisfied by each agent and justify your answer.

The given table compares five task environments using these properties:

- Accessible: The agent can obtain the relevant information about the environment.
- Deterministic: The next state is completely determined by the current state and the agent's action.
- Episodic: Each action or task is independent of previous tasks.
- Static: The environment does not change while the agent is deciding.
- Discrete: The states, percepts and actions are distinct and countable.

Completed table

Agent	Accessible	Deterministic	Episodic	Static	Discrete
Solitaire	No	Yes	Yes	Yes	Yes
Part-Picking Robot	No	No	Yes	No	No
Taxi Driving	No	No	No	No	No
Online Shopping	No	No	No	No	No

Medical Diagnosis	No	No	No	No	No

Justification

1. Solitaire

- Accessible — No: The agent does not have complete access to all relevant information.
- Deterministic — Yes: The outcome of a move is determined by the current state and the chosen move.
- Episodic — Yes: Each decision is treated as an independent episode in the table.
- Static — Yes: The game does not change while the agent is thinking.
- Discrete — Yes: The game has distinct states and moves.

2. Part-Picking Robot

- Accessible — No: The robot may not have complete information about parts and their positions.
- Deterministic — No: Parts may move unpredictably on the conveyor belt.
- Episodic — Yes: Each part-picking task can be treated independently.
- Static — No: The conveyor belt and parts may move while the robot is deciding.
- Discrete — No: The robot's movements and sensor readings can vary continuously.

3. Taxi Driving

- Accessible — No: The driver cannot know every detail of traffic and road conditions.
- Deterministic — No: Other vehicles, pedestrians and traffic conditions are unpredictable.
- Episodic — No: Each driving action affects future decisions.
- Static — No: Traffic and road conditions change over time.
- Discrete — No: Speed, steering and position vary continuously.

4. Online Shopping

- Accessible — No: The agent may not know every detail about sellers, products or stock.
- Deterministic — No: Prices, availability and seller responses may change.
- Episodic — No: Searching, comparing and purchasing are connected decisions.
- Static — No: Listings, prices and availability can change while the agent is deciding.
- Discrete — No: The PDF classifies this environment as non-discrete.

5. Medical Diagnosis

- Accessible — No: The complete internal state of a patient cannot be directly observed.
- Deterministic — No: The same symptoms or treatment may lead to different outcomes.

- Episodic — No: Diagnosis and treatment decisions influence later decisions.
- Static — No: The patient's condition may change over time.
- Discrete — No: Symptoms, measurements and patient conditions can vary continuously.

13. Write pseudocode agent programs for the goal-based and utility-based agents.

(a) Goal-Based Agent

Definition: A goal-based agent selects actions to achieve a specified goal. It considers the current state, the goal and the possible future consequences of its actions.

Pseudocode:

```
function GOAL-BASED-AGENT(percept) returns action

    state ← UPDATE-STATE(state, percept)

    if GOAL-ACHIEVED(state, goal) then
        return NoOp

    actions ← POSSIBLE-ACTIONS(state)

    action ← SEARCH-OR-PLAN(state, goal, actions)

    return action
```

Working:

1. Receives the current percept from the environment.
2. Updates its understanding of the current state.
3. Checks whether the goal has been achieved.
4. If the goal is not achieved, searches or plans for a suitable action.
5. Returns the selected action.

(b) Utility-Based Agent

Definition: A utility-based agent selects the action that maximizes utility. A utility function assigns a numerical value to a state, indicating how desirable that state is.

Pseudocode:

```
function UTILITY-BASED-AGENT(percept) returns action

    state ← UPDATE-STATE(state, percept)

    actions ← POSSIBLE-ACTIONS(state)

    best_action ← NoOp
    max_utility ← -∞

    for each action in actions do
        result ← PREDICT-RESULT(state, action)
        utility ← UTILITY-FUNCTION(result)

        if utility > max_utility then
            max_utility ← utility
            best_action ← action

    return best_action
```

Working:

1. Receives the current percept and updates the state.
2. Identifies the possible actions.
3. Predicts the outcome of each action.
4. Calculates the utility of each predicted outcome.
5. Selects and returns the action with the highest utility.

Difference between the two agents

Goal-based agent	Utility-based agent
Selects actions to achieve a goal.	Selects actions to maximize utility.
Checks whether a desirable goal can be achieved.	Compares how desirable different outcomes are.

Example: Find a route to a destination.	Example: Find a route balancing time, safety and fuel cost.

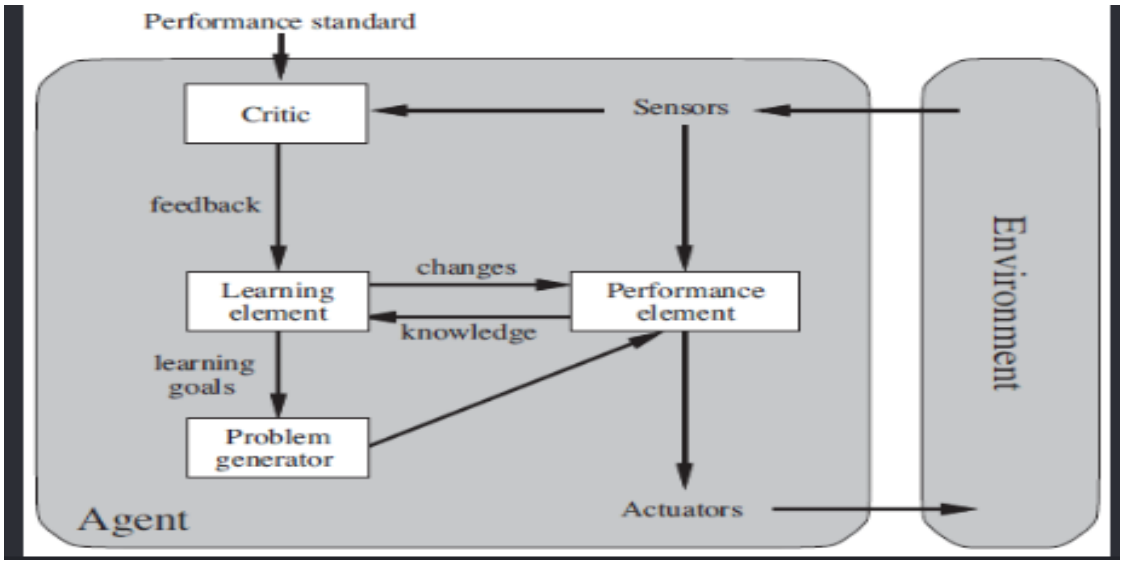
14. With neat diagram explain general learning agent.

Definition

A learning agent is an intelligent agent that improves its performance over time by learning from its experiences and feedback from the environment.

The general learning agent consists of four main components: performance element, learning element, critic, and problem generator.

Diagram



Components of a General Learning Agent

1. Performance Element

- Selects actions based on percepts, internal state and background knowledge.
- Interacts with the environment through sensors and actuators.

2. Learning Element

- Identifies improvements in the agent's performance.
- Uses feedback and experience to modify the performance element.
- Helps the agent adapt and improve over time.

3. Critic

- Evaluates the agent's performance against a given performance standard.
- Provides feedback to the learning element about how well the agent is performing.

4. Problem Generator

- Suggests exploratory actions that may lead to new and useful experiences.
- Helps the agent discover better solutions rather than only repeating known actions.

Working of a Learning Agent

1. The performance element receives percepts from the environment through sensors and selects an action.
2. The action is carried out through actuators, affecting the environment.
3. The critic evaluates the agent's behavior using the performance standard.
4. The critic sends feedback to the learning element.
5. The learning element uses feedback to improve the performance element.
6. The problem generator suggests exploratory actions to gain new knowledge.
7. This process continues, allowing the agent to improve its performance over time.

Example: A self-driving car learns from driving experience and feedback to improve braking, turning and route selection.

Conclusion

A learning agent improves through experience and feedback. The performance element acts, the critic evaluates, the learning element improves, and the problem generator encourages exploration.

15. Mention the different ways to represent states and transitions between them.

A state represents a particular situation of the world, while a transition represents how an action changes one state into another.

The different ways to represent states and transitions are:

1. State-space graph: States are represented as nodes, and transitions between states are represented as directed links or edges labelled with actions.

2. Search tree: The initial state is the root node, and its child nodes represent states reached by applying possible actions. Further transitions are represented by expanding the child nodes.

3. State representation: A state can be represented using a set of variables or properties describing the current situation. For example, a robot's state may include its location and direction.

4. Successor function: It specifies the possible actions and the resulting states for a given state. It can be represented as a collection of action–state pairs.

Example: In route finding, cities represent states and roads represent transitions. Travelling from Bangalore to Mysore is a transition caused by the action of driving.